CUSTOMER SEGMENTATION

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PROJECT VISION

The problem statement focuses on creating a meaningful group of customers and offering relevant products to enhance the overall shopping experience for customers.

The goal of this project is to help businesses improve customer engagement and increase sales by segmenting the customers based on purchasing behavior and recommending products tailored to their individual preferences.

DATASET DESCRIPTION

The dataset used (data.csv) consists of online retail transaction records. Key columns include:

- InvoiceNo (order number)
- StockCode (product code)
- Description (product description)
- Quantity (number of items purchased)
- UnitPrice (price per item)
- CustomerID (customer identifier)
- Country (location)

Total rows: 541909

Target Variable: Customer purchasing behaviour (for segmentation and recommendation)

STAKEHOLDERS

Primary stakeholders:

- Marketing teams looking for customer insights
- Product managers who want better-targeted recommendations
- E-commerce companies that are seeking to optimize customer experience

Secondary stakeholders:

- Customers (benefitting from personalized recommendations
- Data scientists and analysts working on improving models.

INITIAL RESEARCH

Key observations:

- Missing Values: About 25% of entries lacked CustomerID.
- Negative Quantities: Indicated returns or cancellations.

Action Taken:

- Dropped records with missing CustomerID.
- Filtered out transactions with negative Quantity.
- Convert InvoiceDate column to a datetime object.

This ensured only meaningful and clean purchase data was used for modeling.

EXPLORATORY DATA ANALYSIS

You perform visual analysis to understand the distribution of key variables:

- Distribution of UnitPrice: Plot the price distribution using a histogram.
- Distribution of Quantity: Plot the distribution of quantities purchased.
- Scatterplot: Visualize the relationship between Quantity and UnitPrice.
- Boxplot: Show how TotalPrice varies by country.
- Correlation Heatmap: Examine correlations between Quantity, UnitPrice, and TotalPrice.

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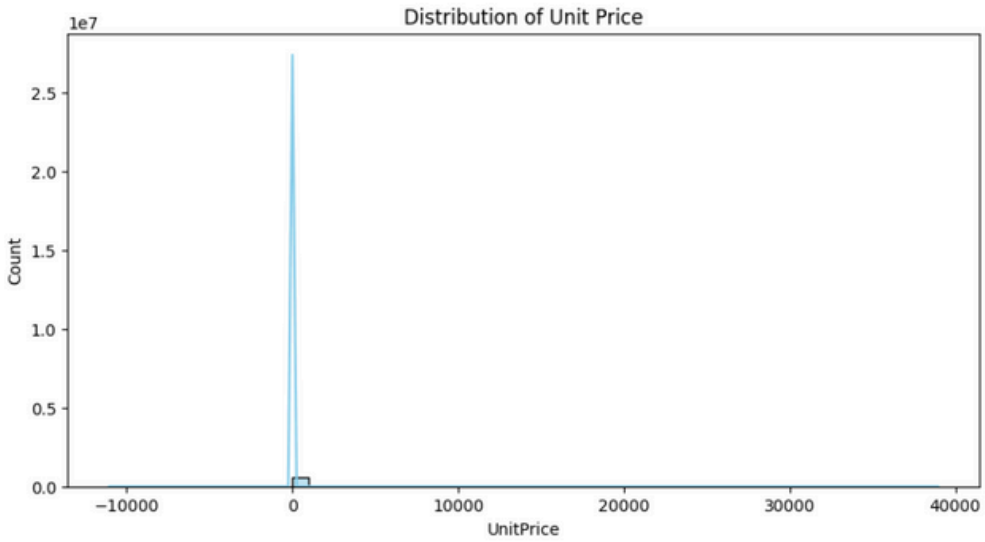
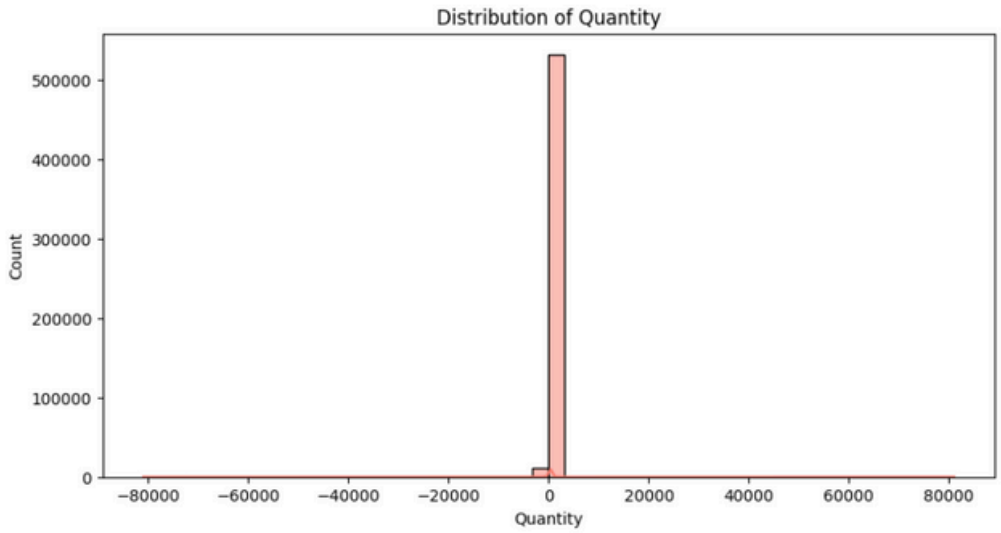
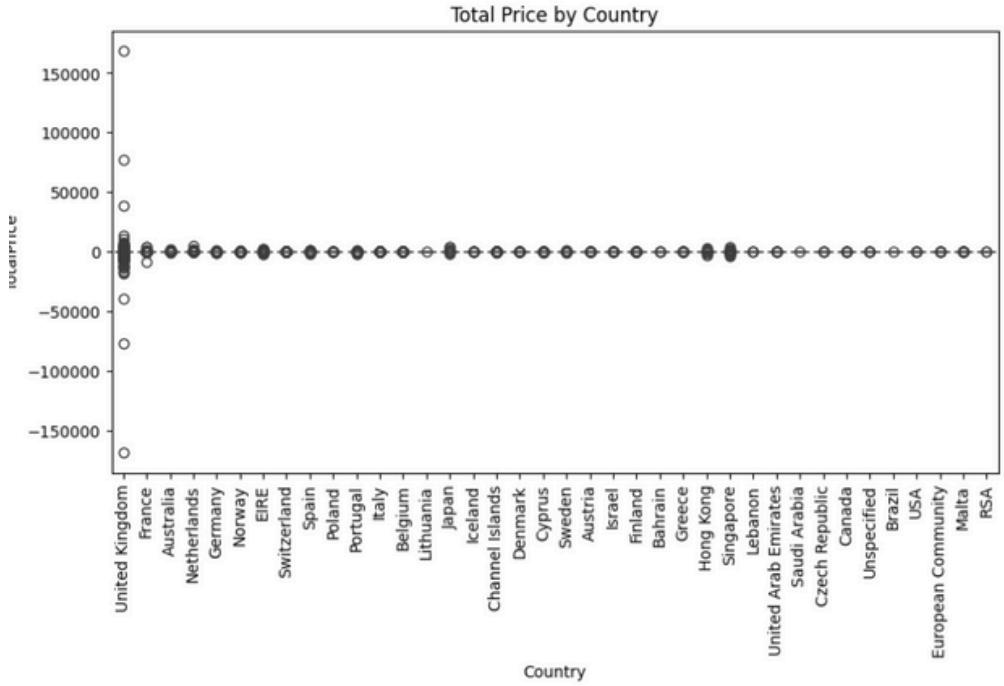
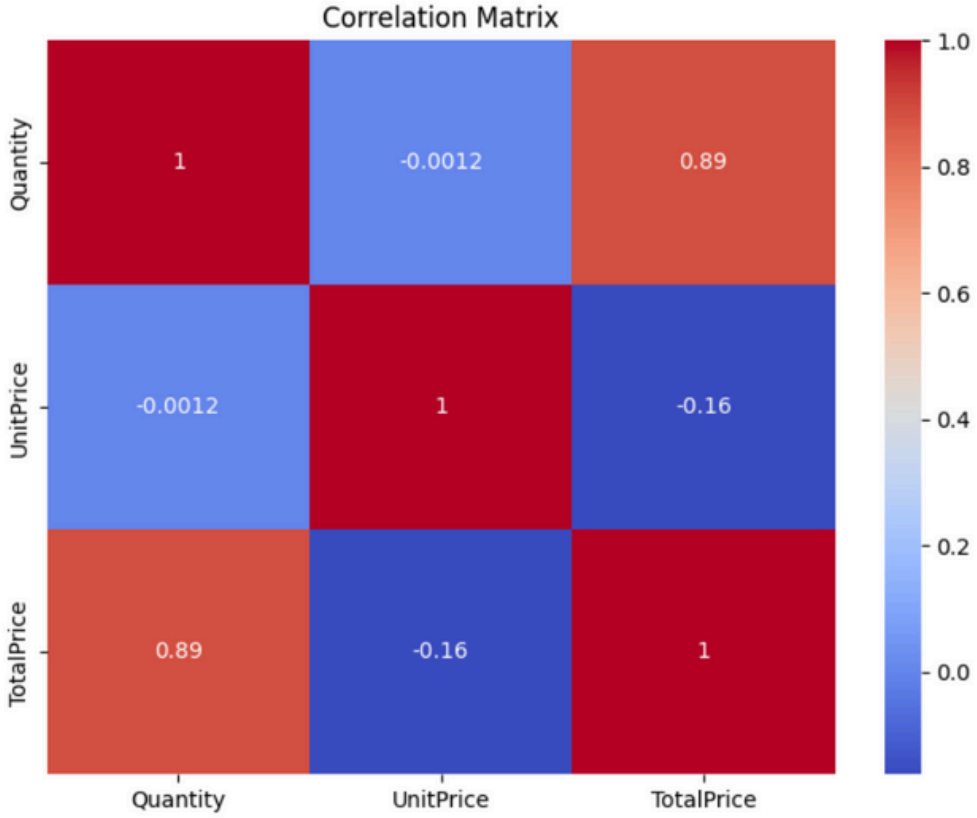

Distribution of Unit Price
1e7
2.5
2.0
1.5
1.0
0.5
0.0
Count
−10000
0
10000
20000
30000
40000
UnitPrice


Distribution of Quantity
500000
400000
300000
200000
100000
0
Count
−80000
−60000
−40000
−20000
0
20000
40000
60000
80000
Quantity

Quantity vs Unit Price
40000
30000
20000
10000
0
−10000
UnitPrice
−80000
−60000
−40000
−20000
0
20000
40000
60000
80000
Quantity

05


Total Price by Country
150000
100000
50000
0
−50000
−100000
−150000
United Kingdom
France
Australia
Netherlands
Germany
Norway
EIRE
Switzerland
Spain
Poland
Portugal
Italy
Belgium
Lithuania
Japan
Iceland
Channel Islands
Denmark
Cyprus
Sweden
Austria
Israel
Finland
Bahrain
Greece
Hong Kong
Singapore
Lebanon
United Arab Emirates
Saudi Arabia
Czech Republic
Canada
Unspecified
Brazil
USA
European Community
Malta
RSA
Country


Correlation Matrix
Quantity
UnitPrice
TotalPrice
1
-0.0012
0.89
-0.0012
1
-0.16
0.89
-0.16
1
Quantity
UnitPrice
TotalPrice
1.0
0.8
0.6
0.4
0.2
0.0

FEATURE ENGINEERING

Key features created:

- TotalPrice: Quantity × UnitPrice
- Recency, Frequency, Monetary (RFM) features:
  - Recency: Days since the last purchase.
  - Frequency: Number of transactions.
  - Monetary: Total spending.

The RFM model was used to characterize customers' behavior for clustering.

CUSTOMER SEGMENTATION

- Apply KMeans clustering to segment customers based on their RFM values into 4 clusters.
- Compute the Silhouette Score to assess the quality of the clustering (ranges from -1 to 1. Here higher values indicate better-defined clusters).

KMeans clustering was chosen because it is simple, efficient, and well-suited for the type of data we had — numerical and continuous RFM (Recency, Frequency, Monetary) features.

Other Method	Why not chosen here
Hierarchical Clustering	Slower for large datasets; very memory-intensive beyond 10,000 rows.
DBSCAN	Good for arbitrary-shaped clusters, but sensitive to parameters like eps and doesn't perform well on high-dimensional data like RFM.
Gaussian Mixture Models	More flexible but assumes data follows Gaussian distributions, and it's computationally heavier.
Spectral Clustering	Very powerful, but not scalable to hundreds of thousands of rows easily without special tricks.

Since this was an exploratory project focused on building a simple, fast, and interpretable model, KMeans was the best balance between performance, interpretability, and computational efficiency.

PRODUCT RECOMMENDATION SYSTEM

- Created a user-item matrix using a pivot table where rows represent customers, columns represent products (StockCode), and values represent the quantity of each product purchased.
- Converted this matrix into a sparse matrix using csr_matrix from Scipy.
- Cosine Similarity: Compute the cosine similarity between customers based on their purchase history. This measures how similar two customers are in terms of products bought.

Recommendation Logic:

- The function recommend_customers() suggests similar customers based on cosine similarity. It finds the top n most similar customers and returns their CustomerIDs.
- The function recommend_products() suggests products that similar customers have bought but the target customer has not yet purchased. It ensures that recommendations are based on products the customer hasn't already bought.

Why was Cosine Similarity used for the project ?

Reason	Explanation
No product features needed .	We didn’t need to know product types, categories, or any other metadata. Just purchase history!
Works well with only purchase data . .	Since the dataset only has CustomerID, StockCode, Quantity, and Description, collaborative filtering fits perfectly.
Simple and fast .	Cosine similarity is quick to calculate even for big datasets.
Personalized .	Recommendations are based on actual customer behavior, not general trends.

STATISTICAL INSIGHTS

- Cluster 0: High spenders, frequent buyers (most loyal customers).
- Cluster 1: Occasional buyers with moderate spending.
- Cluster 2: Infrequent and low-value buyers.
- Cluster 3: At risk of losing

Silhouette Score: Provided validation for cluster quality (~0.64).

HYPERPARAMETER TUNING

Iterations: Multiple KMeans runs with varying k (from 2 to 10) to select the best k value for our model

Evaluation:

- Elbow curve analysis.
- Silhouette score analysis.

REAL-WORLD IMPACT

- For Businesses: Helps in creating targeted marketing campaigns (loyalty rewards, discounts).
- For Customers: Enables highly personalized shopping experiences.
- For Product Strategy: Assists in inventory planning based on cluster behavior.

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DEPLOYMENT

```python
# Take input from user
customer_id_input = float(input("Enter Customer ID: "))

# Get output
recommendations = recommend_customers(customer_id_input)

# Show output
print(recommendations)
```

```
Enter Customer ID:  12347.0
Index([14326.0, 12355.0, 14257.0, 13532.0, 15539.0], dtype='float64', name='CustomerID')
```

```python
# Take input from user
customer_id = float(input("Enter Customer ID: "))

# Get product recommendations
product_recommendations = recommend_products(customer_id)

# Show output
print("Recommended products for the customer:")
print(product_recommendations)
```

```
Enter Customer ID:  12347.0
Recommended products for the customer:
['SET/3 OCEAN SCENT CANDLE JEWEL BOX', 'ANTIQUE SILVER T-LIGHT GLASS', 'ENAMEL FLOWER JUG CREAM',
'FELTCRAFT 6 FLOWER FRIENDS', 'HANGING HEART JAR T-LIGHT HOLDER']
```